Lecture 18 - Economics and Business Administration
Job Order Costing Systems

Agenda

- Job Order Costing Systems — The job cost sheet and source documents.
- Allocation of indirect costs to jobs
- Cost flow in job order costing systems and journal entries
- Accounting treatment of over/under-absorption of manufacturing overhead

Cost Measurement Systems

- **Job Order Costing Systems**

  Used for products or services immediately identified as individual units (unique items) or small batches, produced to customer specifications through multi-stage transformation processes whose sequence may vary from product to product.

  Examples: ships, catalogues, consulting engagements, etc.

  Method: cost accumulation by job.

- **Process Costing Systems**

  Used for products or services that are not separately identifiable, produced in large quantities through continuous-flow transformation processes with fixed sequential stages.

  Examples: oil, cement, paint, etc.

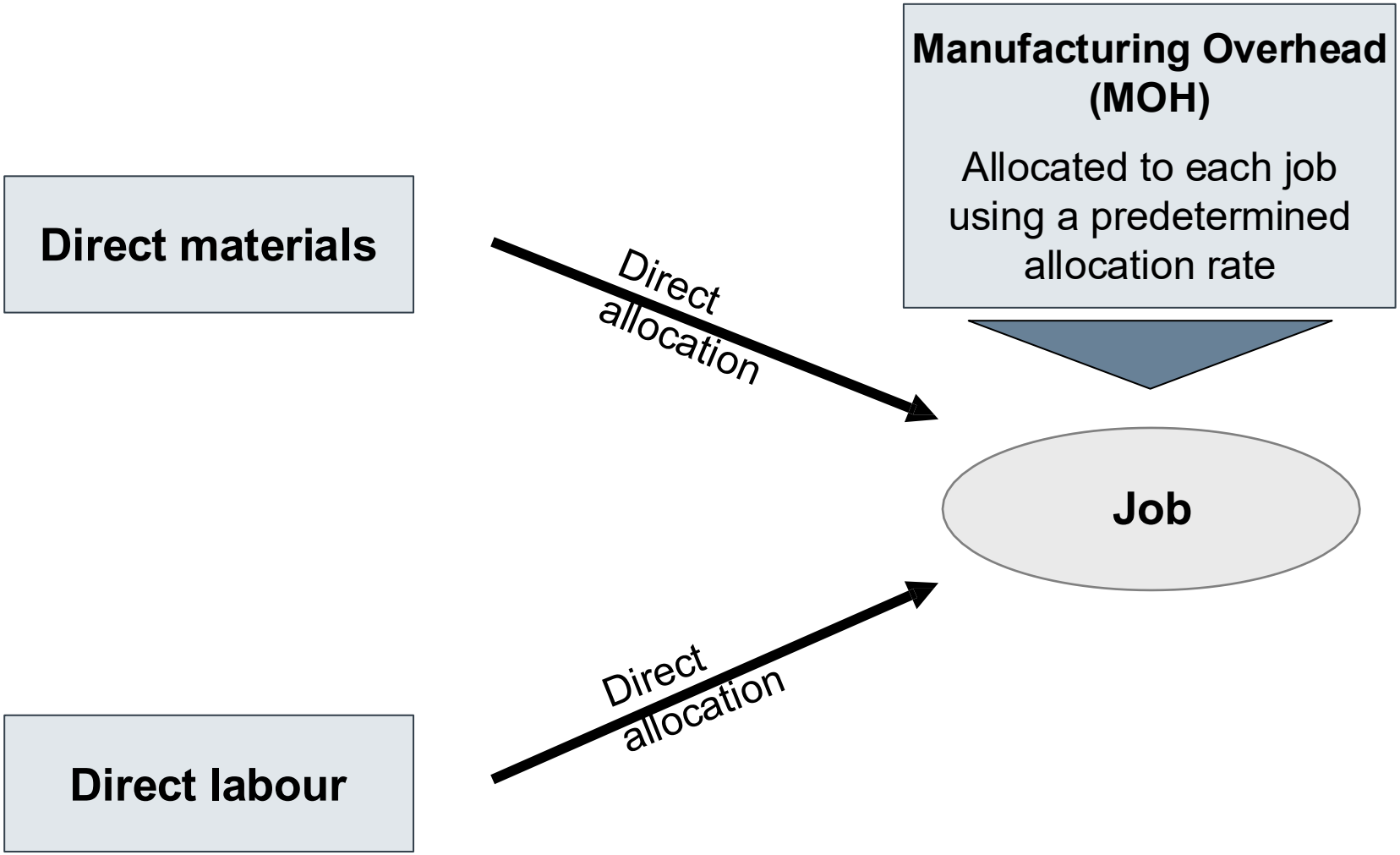
  Method: cost accumulation by department (or process stage)

Job Order Costing Systems

Job Order | Process

- ✓ Many different products or small batches produced each period
- ✓ Products are made to order
- ✓ Costs are traced or allocated to jobs
- ✓ Separate cost records must be maintained for each distinct product or job

Job Order Cost Measurement



Allocation of Manufacturing Overhead

The Allocation Rate (bdg) used to allocate manufacturing overhead to jobs is determined at the start of the period.

$$\text{AR (bdg)} = \frac{\text{Budgeted manufacturing overhead}}{\text{Budgeted volume of the cost driver}}$$

the intensity of absorption of manufacturing overhead (indirect costs)

Allocation of Manufacturing Overhead

Based on estimates, determined before the start of the period

Allocated Manufacturing Overhead = AR bdg x Actual level of the Cost Driver

Actual level of the cost driver

such as units produced, direct labour hours or machine hours for the period.

Overhead Allocation Example

- ONAMA allocates overhead based on direct labour hours.
- Total budgeted manufacturing overhead for the year is €640,000.
- Total budgeted labour cost is €1,400,000 and total budgeted labour hours are 160,000.

What is ONAMA's budgeted hourly rate (AR bdg)?

Overhead Allocation Example

$$\text{AR bdg} = \frac{\text{Total budgeted manufacturing overhead for the next period}}{\text{Budgeted volume of the cost driver for the next period}}$$

$$\text{AR bdg} = \frac{€640{,}000}{160{,}000 \text{ direct labour hours (DL hrs)}}$$

= €4.00 per DL hr

For every direct labour hour worked on a job, €4.00 of manufacturing overhead is allocated to that job.

Job Cost Sheet

Manufacturing overhead allocated to the job using a predetermined overhead rate of €4 per direct labour hour (DL hr).

ONAMA — Job Cost Sheet

Job No.		Start date	Completion date	Department	Item
MA 54-09-30		4-3-01	3-5-01	B3	Wooden goods crate

Direct Materials		Direct Labour			Manufacturing Overhead	
Req. No.	Amount	Time Ticket	Hours	Amount	Hours	Rate
X7-6890	€ 116	36	8	€ 88	8	€ 4

Cost summary			Units Shipped
Direct Materials	€ 116		
Direct Labour	€ 88		
Manufacturing Overhead	€ 32		
Total Cost	**€ 236**		
Unit product cost	**€ 118**		

Analysis of Cost Flows and Summary Journal Entries for a Job Order Costing System

Cost Flow – Raw Materials Purchase (Entry 1)

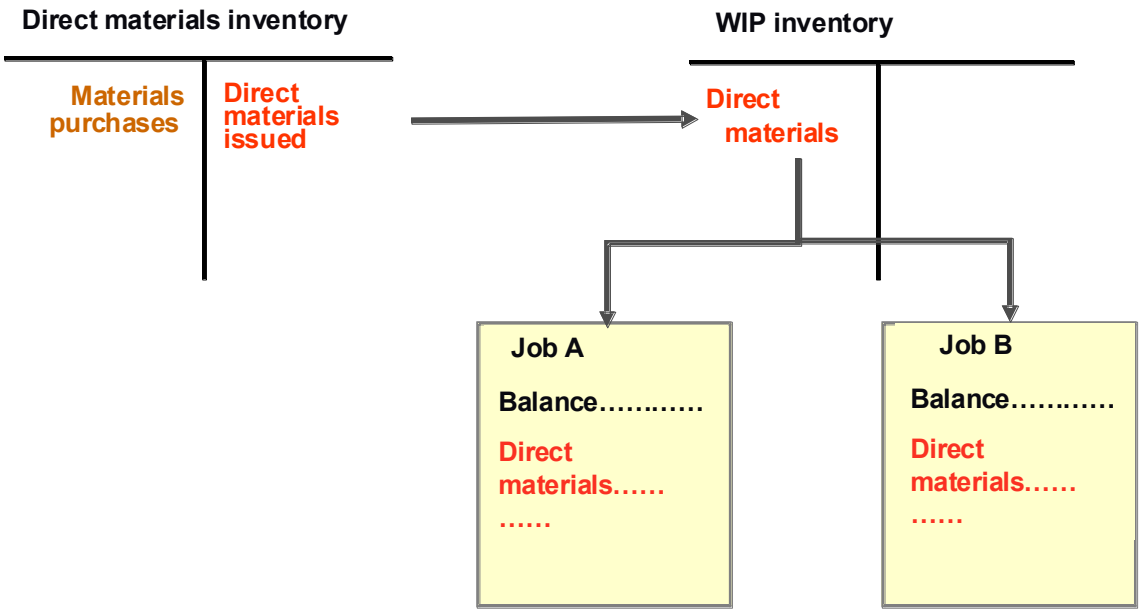
Direct materials purchased

Date	Description	Amount	Debit	Credit
	Raw Materials Inventory		XXXXX	
	Account payables			XXXXX

Cost Flow – Materials Usage (Entry 2)

Direct materials issued to a job increase WIP inventory and reduce direct materials inventory

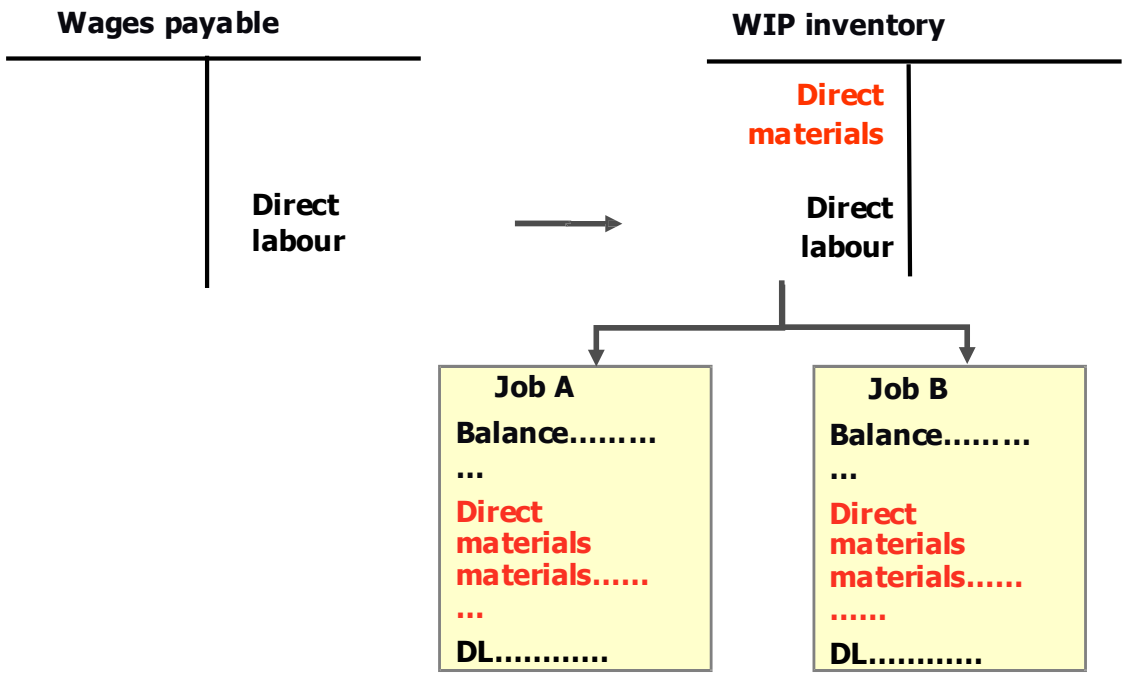
Date	Description	Amount	Debit	Credit
	WIP Inventory		XXXXX	
	Direct Materials Inventory			XXXXX



Cost Flow – Labour (Entry 3)

Direct labour cost incurred increases WIP inventory

Date	Description	Amount	Debit	Credit
	WIP Inventory		XXXXX	
	Wages Payable			XXXXX



Cost Flow – Actual Overhead Costs (Entry 4a)

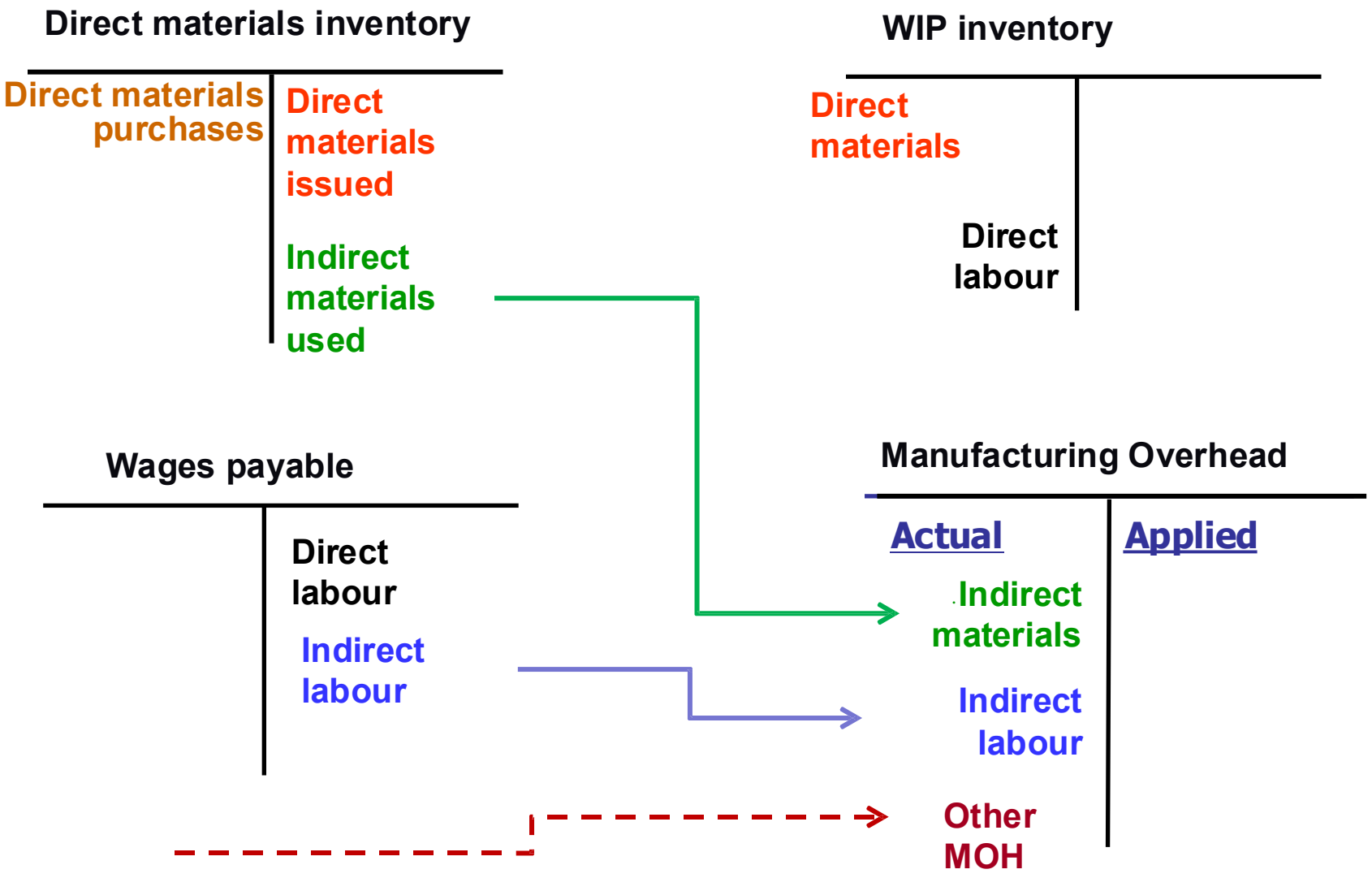
Indirect costs are recorded in the Manufacturing Overhead account when actually incurred

Date	Description	Amount	Debit	Credit
	Manufacturing Overhead		XXXXX	
	Accounts Payable (indirect materials)			XXXXX
	Wages Payable (indirect labour)			XXXXX
	Insurance Payable			XXXXX
	Accumulated Depreciation			XXXXX

In practice, setting aside the example, actual indirect costs are recorded with specific entries for each type (electricity, consumables, etc.) at different times.

Recording of actual costs can (and usually does) occur after the allocation phase using the AR bdg

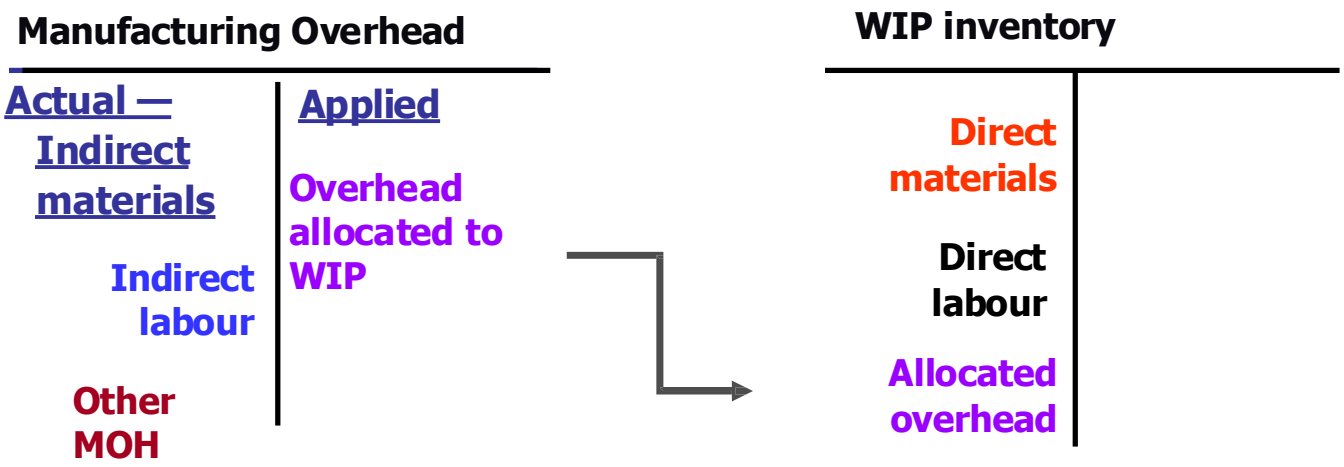
Cost Flow in Job Order Costing Systems



Cost Flow – Applied Overhead (Entry 4b)

WIP inventory increases when manufacturing overhead is allocated to jobs

Date	Description	Amount	Debit	Credit
	WIP Inventory		XXXXX	
	Manufacturing Overhead			XXXXX



If actual and applied manufacturing overhead differ, an adjustment is required at year end.

Cost Flow – Cost of Goods Manufactured (Entry 5)

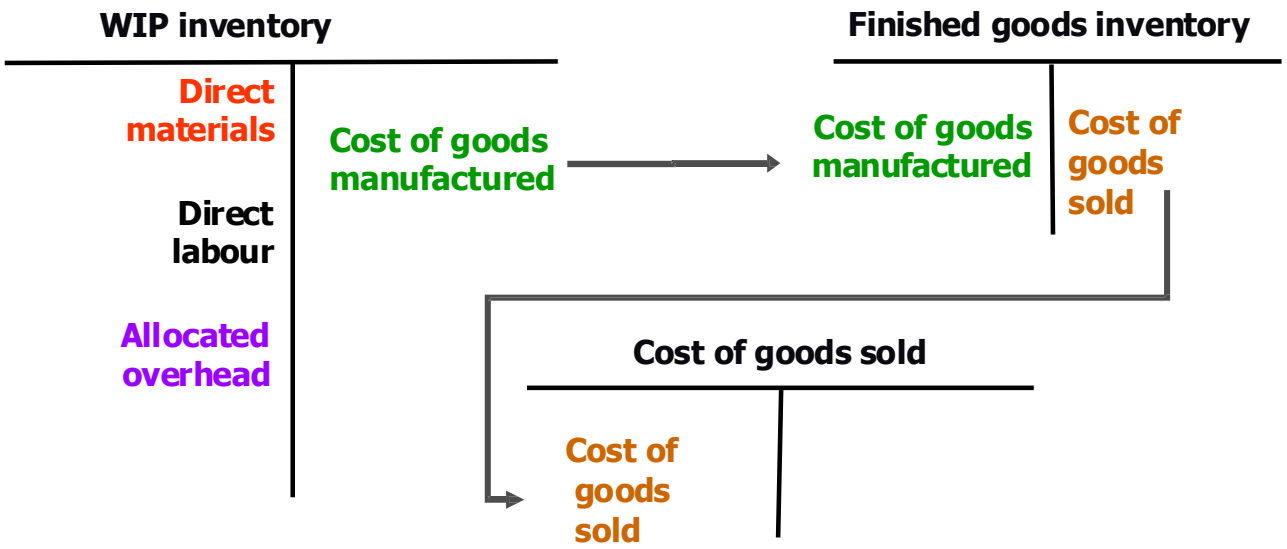
When jobs are completed, the cost of goods manufactured is transferred from WIP inventory to finished goods inventory

Date	Description	Amount	Debit	Credit
	Finished Goods Inventory		XXXXX	
	WIP Inventory			XXXXX

Cost Flow – Sales and Cost of Goods Sold (Entries 6a and 6b)

When finished goods are sold, two entries are required: (1) to record the sale and (2) to record the cost of goods sold and reduce finished goods inventory

Date	Description	Amount	Debit	Credit
	Accounts Receivable		XXXXX	
	Sales Revenue			XXXXX
	Cost of Goods Sold		XXXXX	
	Finished Goods Inventory			XXXXX

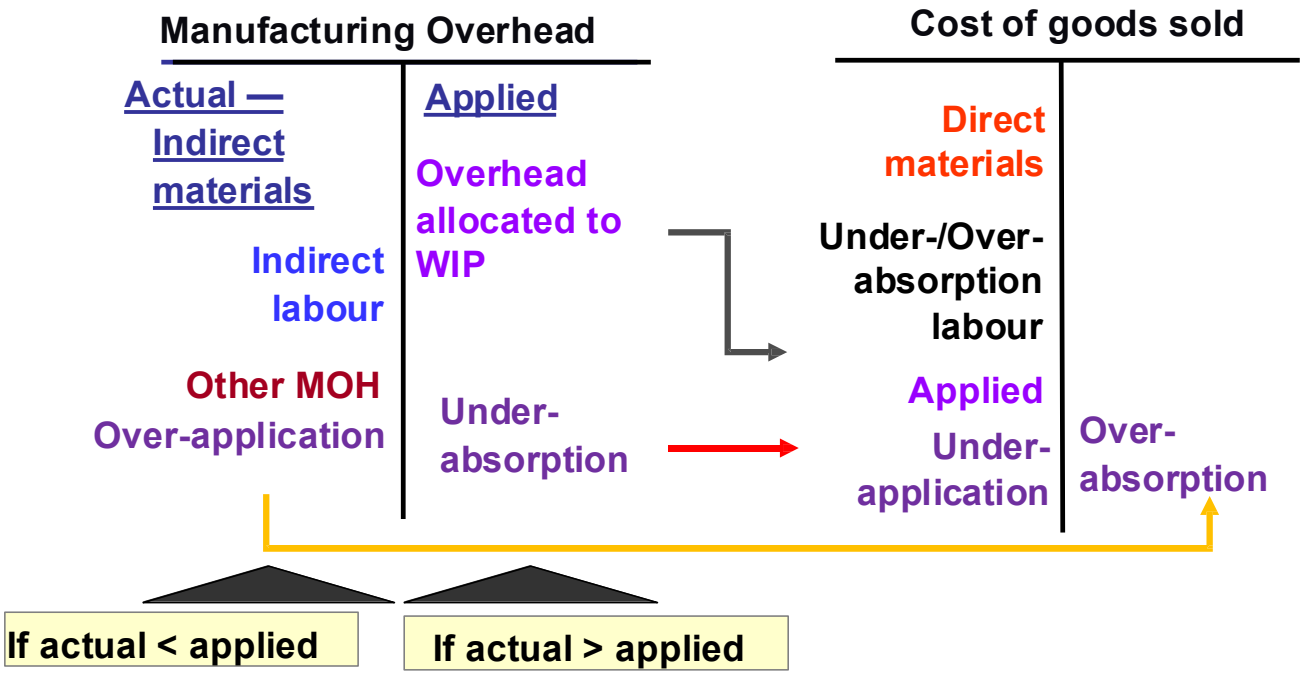


Cost Flow – Over/Under-Absorption of Manufacturing Overhead (Entry 7)

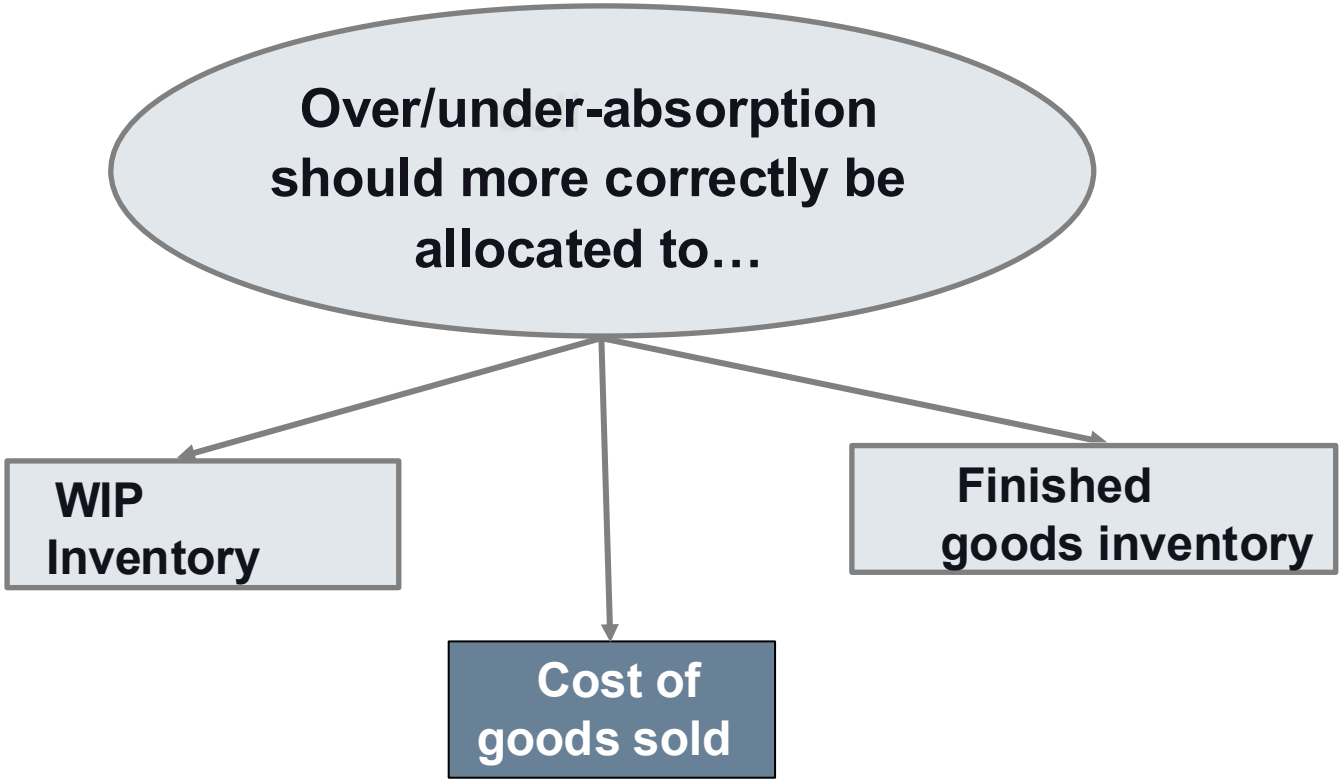
At year end, the over/under-absorption of manufacturing overhead is closed out.

If under-absorbed

Date	Description	Amount	Debit	Credit
	Cost of Goods Sold		XXXXX	
	Manufacturing Overhead			XXXXX



Alternative Treatments for Over/Under Application of Manufacturing Overhead



Exercise

Case: Alfa

Case Text (1/2)

- Alfa S.p.A. has two jobs in progress during the month of October:
- At end of September, production costs of €30,000 had been recorded for Job A.
- On 1 October, Alfa had €5,000 of direct materials in inventory. During the month it purchased €50,000 more, while €45,000 of materials were issued to production. Job A used €15,000 of direct materials and Job B used €30,000.
- In October, direct labour cost was €60,000, of which €20,000 relates to Job A and €40,000 to Job B.

- Manufacturing overhead in October totalled €94,000 (comprising: indirect materials €4,000, indirect labour €10,000, utilities and rentals €20,000, taxes €10,000, insurance €5,000, machinery depreciation €45,000).

Case Text (2/2)

- Alfa uses machine hours for the allocation rate (bdg), which is €5 per machine hour.
- In October, Job A used 8,000 machine hours and Job B used 10,000 hours.
- Job A was completed in October. During the same period, 800 units were delivered to customers, generating revenue of €200,000.
- In October: selling and administrative salaries €40,000; advertising €35,000; other selling and administrative expenses €10,000; office equipment depreciation €5,000.

Question

Prepare the accounting flow, recording entries in the single accounts and journal, noting that Alfa closes any under/over-absorption of manufacturing overhead to cost of goods sold.

Cost Flow – Materials Purchases (Entry 1)

Direct materials purchases are recorded in an inventory account

GENERAL JOURNAL

Date	Description	Debit	Credit
	Raw Materials Inventory	50.000	
	Accounts Payable		50.000

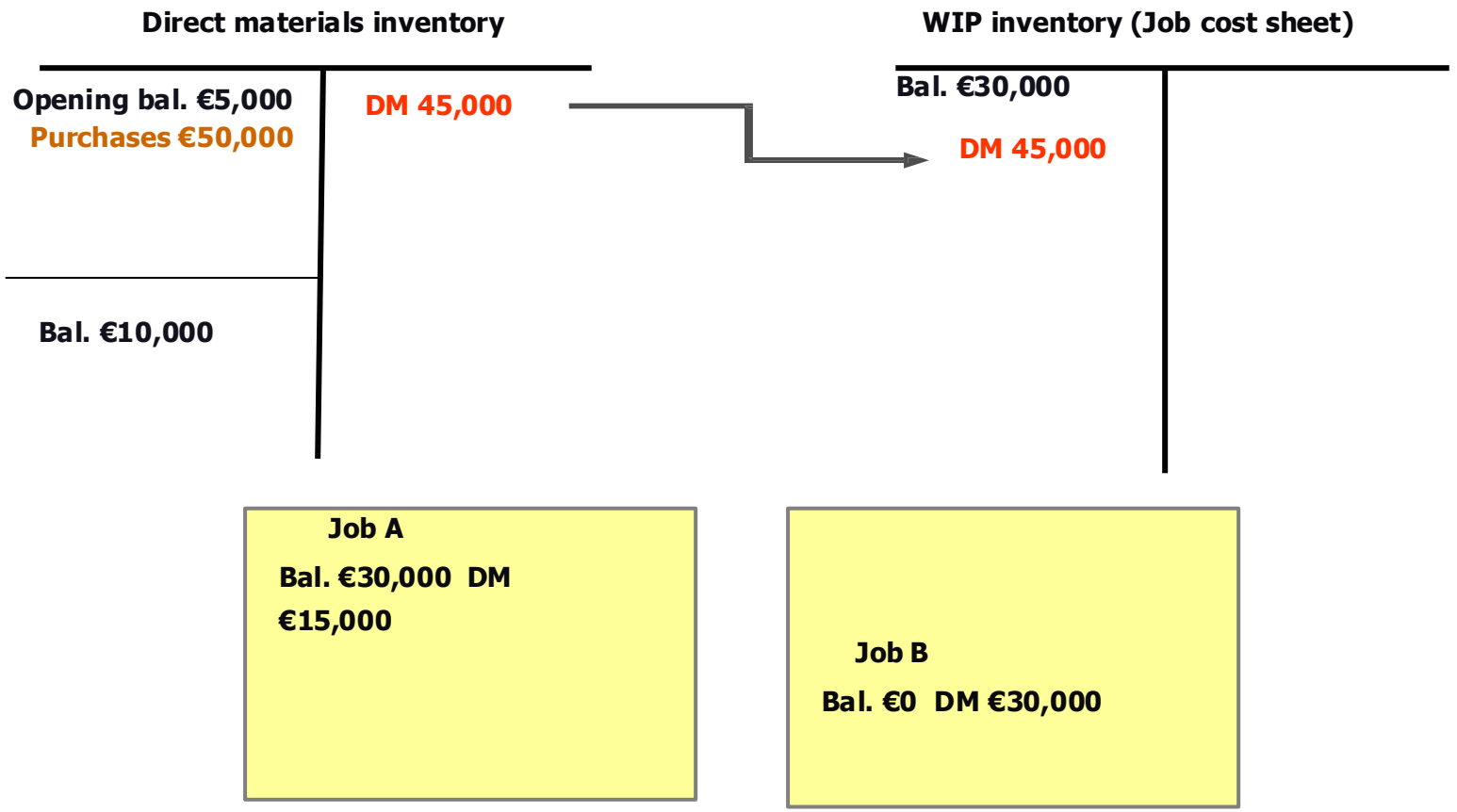
Cost Flow – Materials Usage (Entry 2)

Direct materials issued to a job increase WIP inventory and reduce direct materials inventory

GENERAL JOURNAL

Date	Description	Debit	Credit
	WIP Inventory	45.000	
	Raw Materials Inventory		45.000

Cost Flow in Job Order Costing Systems

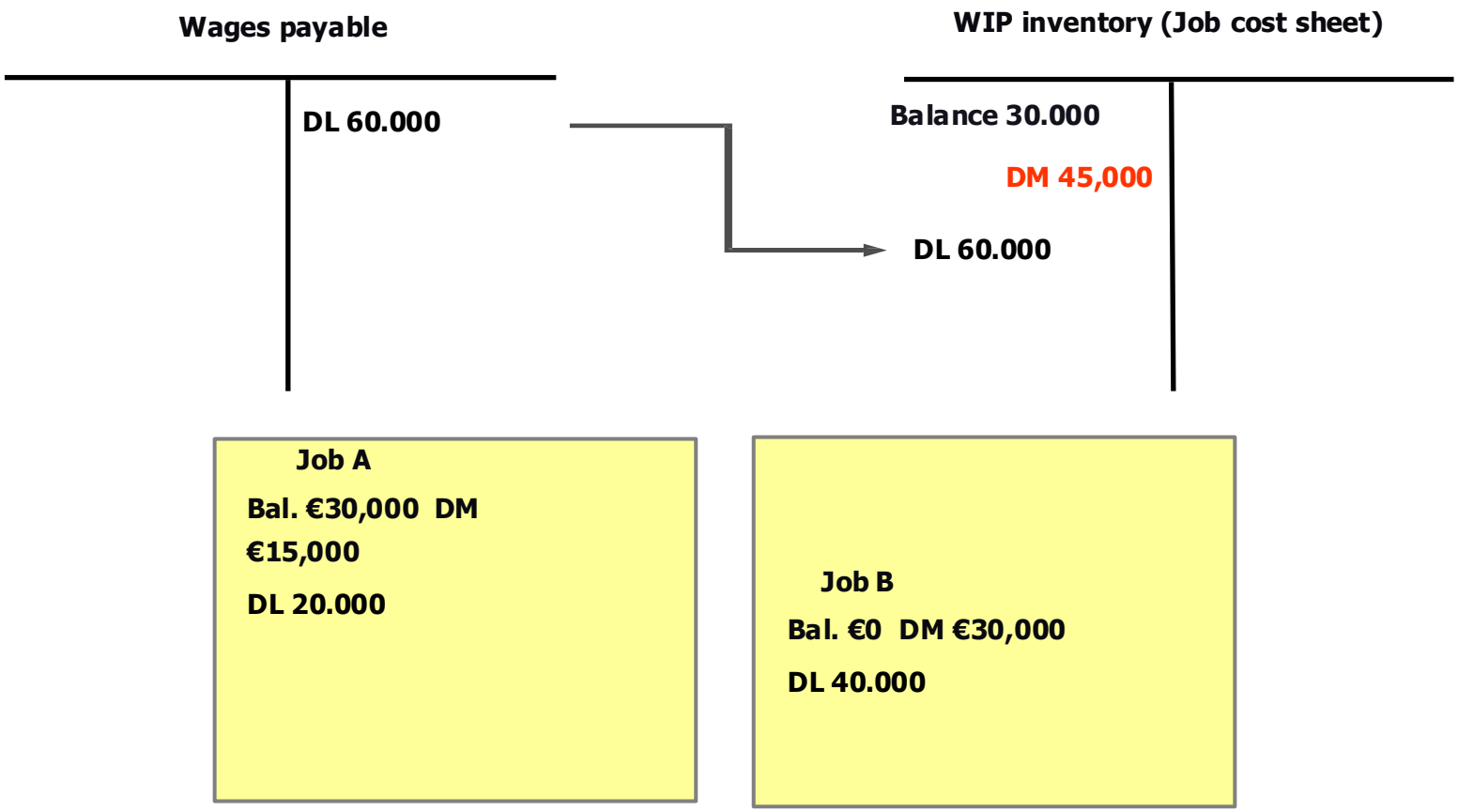


Cost Flow – Labour (Entry 3)

Direct labour cost incurred increases WIP inventory

Date		Description	Debit	Credit
		WIP Inventory	60.000	
		Wages Payable		60.000

Cost Flow in Job Order Costing Systems



Cost Flow – Actual Overhead Costs (Entry 4a)

In addition to indirect materials and indirect labour, other manufacturing overhead costs are charged to “Manufacturing Overhead” when incurred.

Date		Description	Debit	Credit
		Accounts Payable (indirect materials)		4.000
		Wages Payable (indirect labour)		10.000
		Accts Pay. (services & other prod. costs)		20.000
		Tax Payable		10.000
		Insurance Payable (premiums)		5.000
		Accumulated Depreciation		45.000

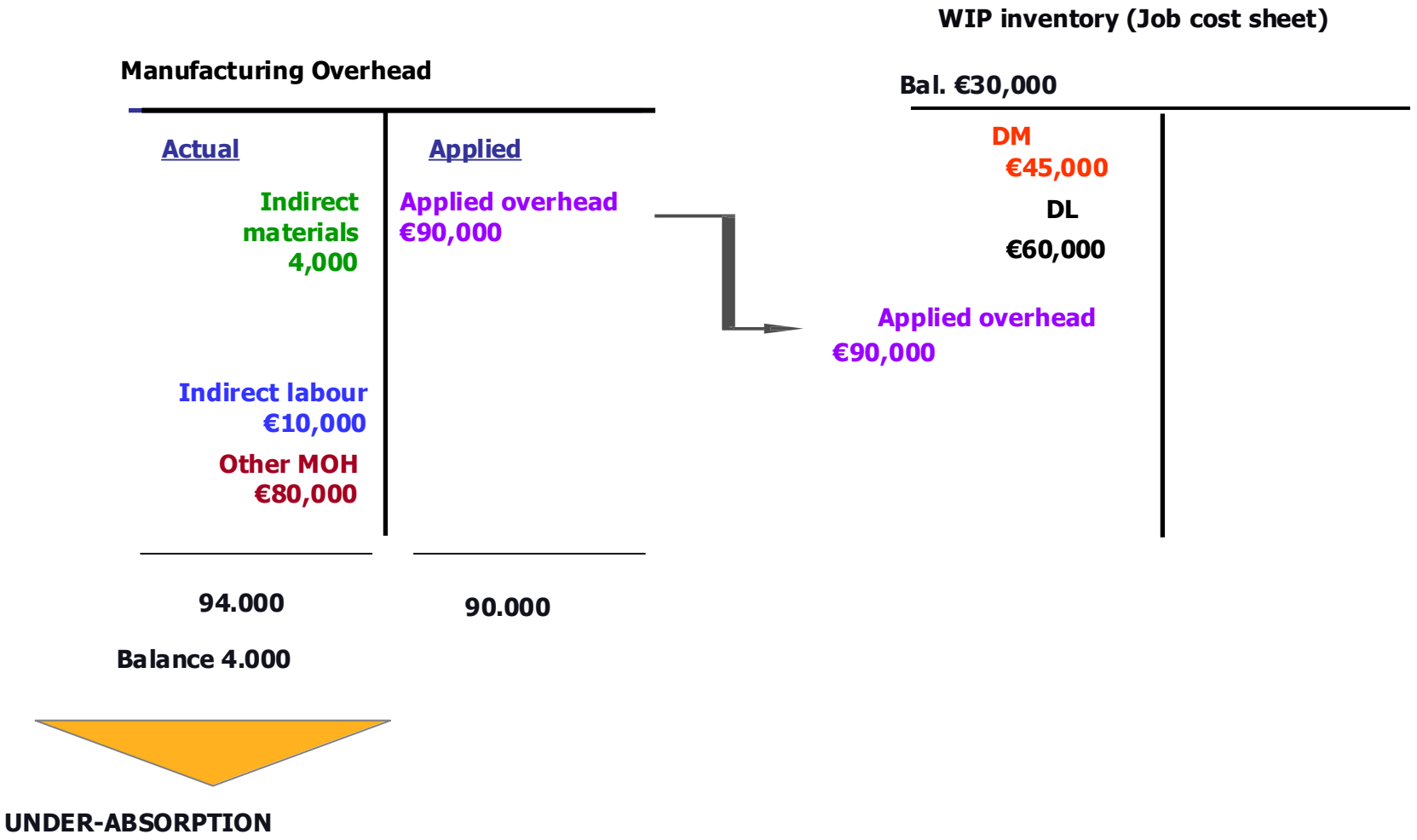
Cost Flow – Applied Overhead (Entry 4b)

WIP inventory increases when manufacturing overhead is allocated to jobs

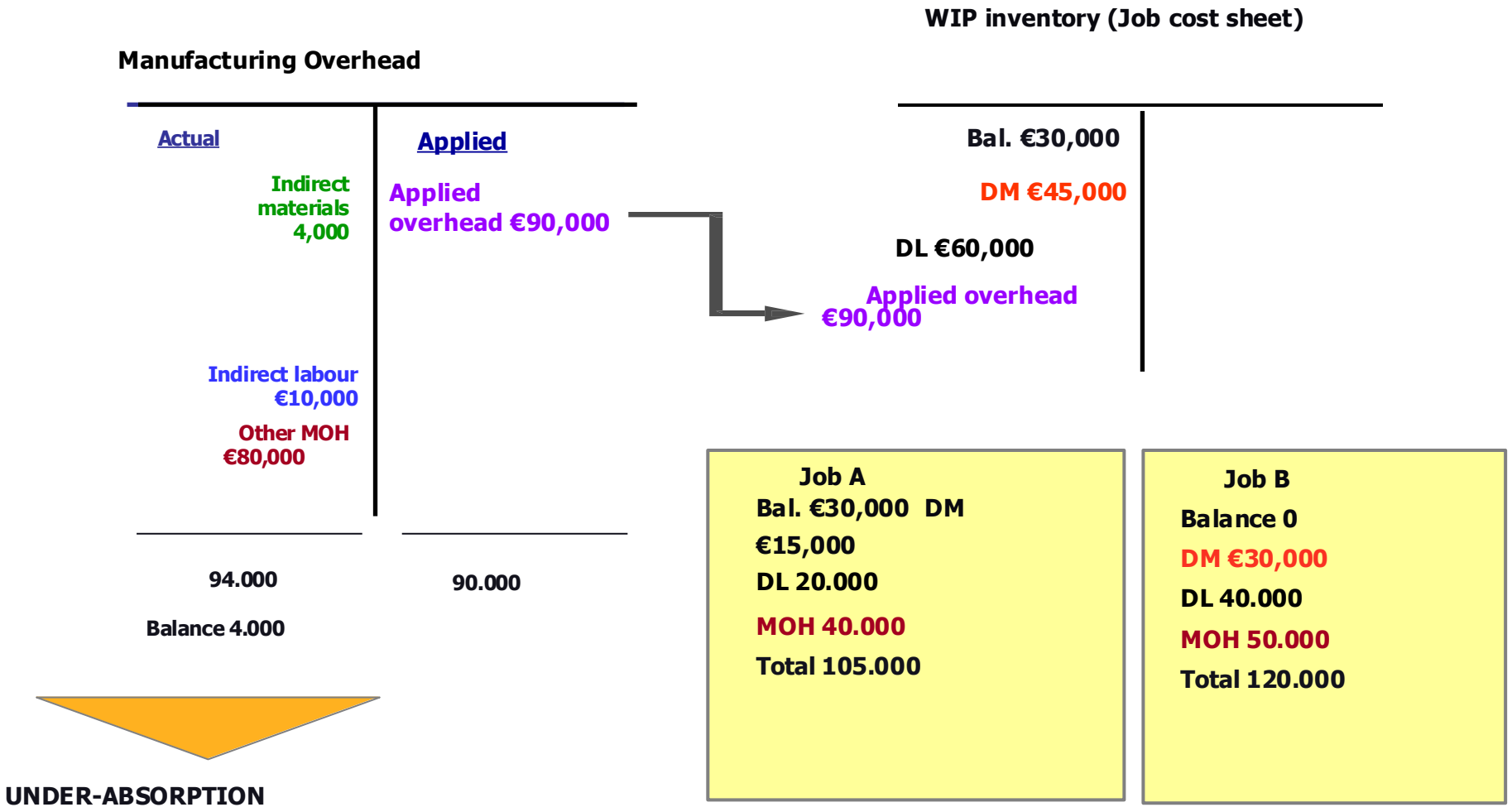
GENERAL JOURNAL

Date		Description	Debit	Credit
		WIP Inventory	90.000	
		Manufacturing Overhead		90.000

Cost Flow in Job Order Costing Systems



Cost Flow in Job Order Costing Systems



Cost Flow – Cost of Goods Manufactured (Entry 5)

When jobs are completed, the cost of goods manufactured is transferred from WIP inventory to Finished Goods inventory.

GENERAL JOURNAL

Date	Description	Debit	Credit
	Finished Goods Inventory	105.000	
	WIP Inventory		105.000

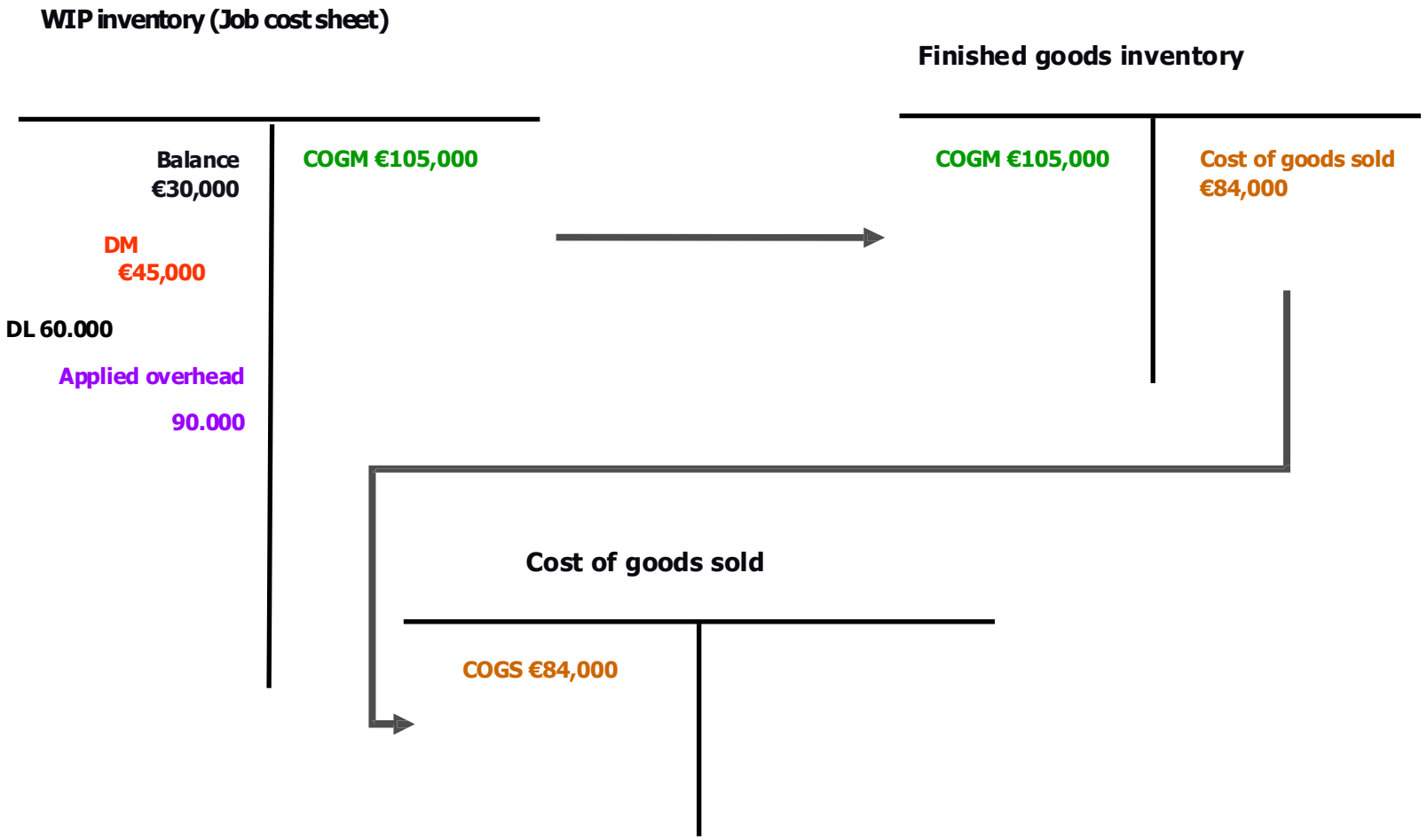
Cost Flow – Sales and Cost of Goods Sold (Entries 6a and 6b)

When finished goods are sold, two entries are required: (1) to record the sale and (2) to record the cost of goods sold and reduce finished goods inventory.

GENERAL JOURNAL

Date	Description	Debit	Credit
	Accounts Receivable	200.000	
	Sales Revenue		200.000
	Cost of Goods Sold	84.000	
	Finished Goods Inventory		84.000

Cost Flow in Job Order Costing Systems



Cost Flow – Period Costs

Non-production costs (period costs) are charged to the Income Statement when incurred.

Date	Description	Debit	Credit
	Selling & Admin. Salaries	40.000	
	Wages Payable		40.000
	Depreciation Expense	5.000	
	Accumulated Depreciation		5.000
	Advertising Costs	35.000	
	Other Selling & Admin.	10.000	
	Accounts Payable		45.000

Cost of Goods Manufactured

Direct materials		
Opening direct materials inventory	€ 5.000	
+ Direct materials purchases	€ 50.000	
Total direct materials available	**€ 55.000**	
– Closing direct materials inventory	€ 10.000	
Direct materials		€ 45.000
Direct labour		€ 60.000
Manufacturing overhead allocated to WIP		€ 90.000
Total production costs		**€ 195.000**
+ Opening WIP inventory		€ 30.000
– Closing WIP inventory		€ 120.000
Cost of goods manufactured		**€ 105.000**

Cost of Goods Sold

Cost of Goods Sold	
Opening finished goods inventory	€ 0
+ Cost of goods manufactured	€ 105.000
Cost of goods available for sale	**€ 105.000**
– Closing finished goods inventory	€ 21.000
Unadjusted COGS	**€ 84.000**
+ Under-absorbed overhead	€ 4.000
Adjusted COGS	**€ 88.000**

October Income Statement

Sales		**€ 200.000**
Cost of Goods Sold (adjusted)		**€ 88.000**
Gross margin		**€ 112.000**
Selling and administrative costs		€ 90.000
Salaries	€ 40.000	
Advertising costs	€ 35.000	
Other costs	€ 10.000	
Depreciation charge	€ 5.000	
Net operating income		**€ 22.000**

Motorboat Exercise

LuxuryYacht produces medium and large motorboats and uses a normal costing job order system. The Predetermined Overhead Rate (POR) is based on DL cost.

The following information applies to June:

- The company has 4 jobs in progress (2 started in May and 2 in June itself)
- The company allocates manufacturing overhead to jobs at month end (or upon job completion if earlier)
- In June, the company completes Jobs Isabella, Federica, and Giovanna
- Also in June, it sells Jobs Federica and Giovanna at prices of €1,200,000 and €480,000 respectively

The data recorded by the cost measurement system for the jobs (before June overhead allocation) are as follows:

	Isabella		Federica		Giovanna		Jessica	
	May	June	May	June	May	June	May	June
Direct materials costs	48,000	25,000	68,000	48,000		60,000		57,000
DL costs	125,000	60,000	160,000	40,000		140,000		85,000
Manufacturing overhead allocated	75,000		96,000					

Questions

1. Determine the opening WIP inventory value for June
2. Determine the Allocation Rate (bdg)
3. Determine the manufacturing overhead allocated to jobs in June
4. Determine the value of transfers to finished goods inventory in June
5. Determine the closing WIP inventory value for June
6. Determine the closing finished goods inventory for June (no opening balance)
7. Determine the cost of goods sold for June
8. Knowing that manufacturing overhead in June amounted to €205,000, determine the over/under-absorption for the month
9. Complete the ledger accounts below for June, indicating known opening and closing balances
10. Knowing that non-production costs in June were €300,000, prepare the monthly income statement

Solutions (1/4)

Opening WIP inventory

Costs recorded in May

Isabella	248,000
Federica	324,000
	572,000

Overhead allocated in June

Isabella	36,000
Federica	24,000
Giovanna	84,000
Jessica	51,000
	195,000

AR bdg (€/€DL)

Normal costing: AR is constant across months, based on DL cost

The AR bdg is derived from May allocated values relative to DL cost recorded (e.g. Isabella = 75,000/125,000)

60,000 × 0.6

Accounting position post-

	Isabella		Federica		Giovanna		Jessica	
	May	June	May	June	May	June	May	June
Direct materials costs	48,000	25,000	68,000	48,000		60,000		57,000
DL costs	125,000	60,000	160,000	40,000		140,000		85,000
Manufacturing overhead allocated	75,000	36,000	96,000	24,000		84,000		51,000
Tot	**248,000**	**121,000**	**324,000**	**112,000**	**-**	**284,000**	**-**	**193,000**

Solutions (2/4)

Value of transfers to FG inventory

The transfer includes all costs recorded on completed jobs

Isabella	369,000
Federica	436,000
Giovanna	284,000
	1,089,000

Closing WIP inventory

Jessica	193,000

Closing FG inventory

Isabella	369,000

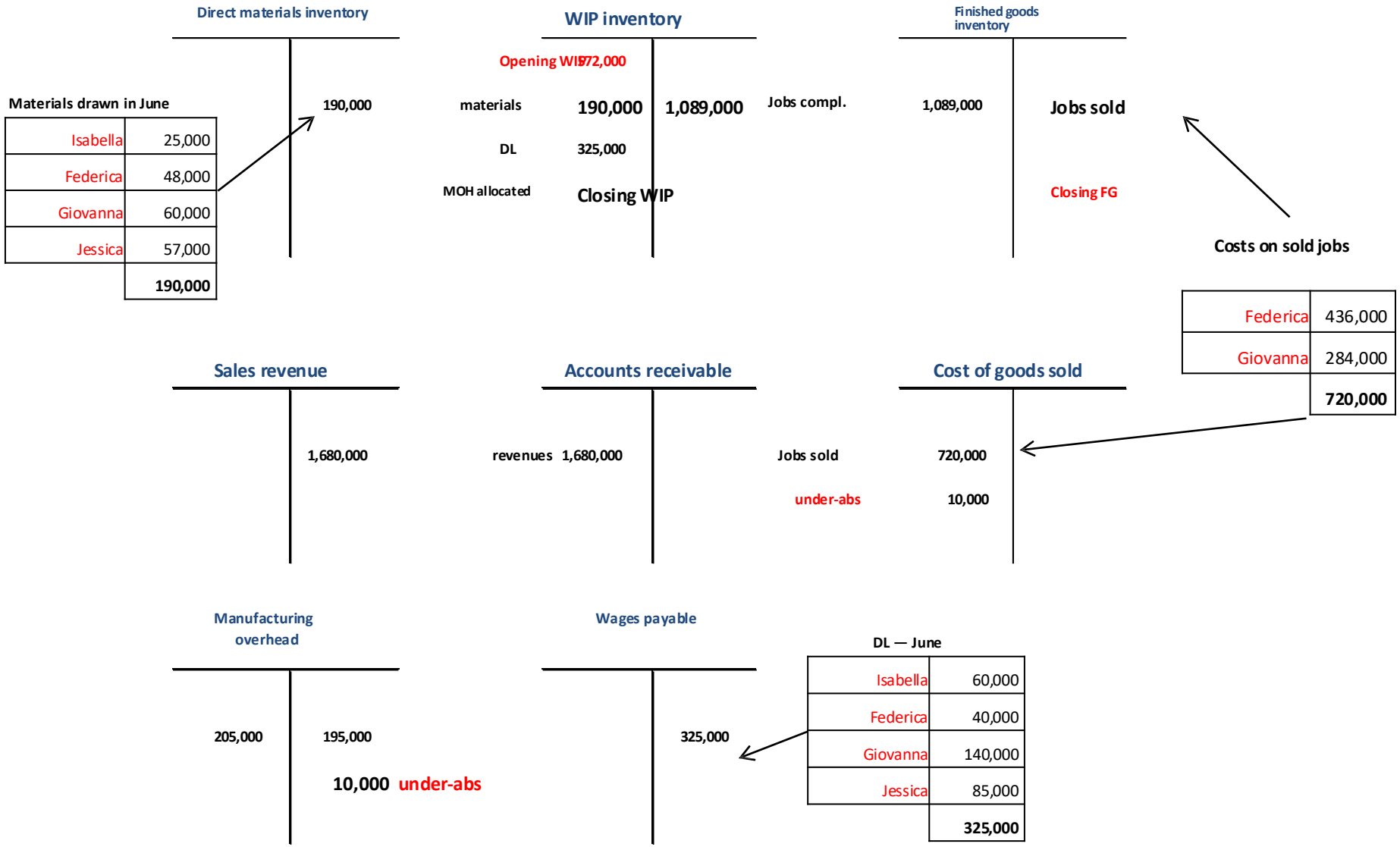
Cost of goods sold

Federica	436,000
Giovanna	284,000
	720,000

Over/Under-absorption

Overhead allocated		195,000
Actual overhead	€	205.000
UNDER-ABSORPTION	**€**	**10.000**

Solutions (3/4)



Solutions (4/4)

10	Income Statement — JUNE	
	Revenues	€ 1.680.000
	Adjusted COGS	€730,000
	Gross Margin	**€ 950.000**
	Non-production costs	€ 300.000
	Operating Income	**€ 650.000**